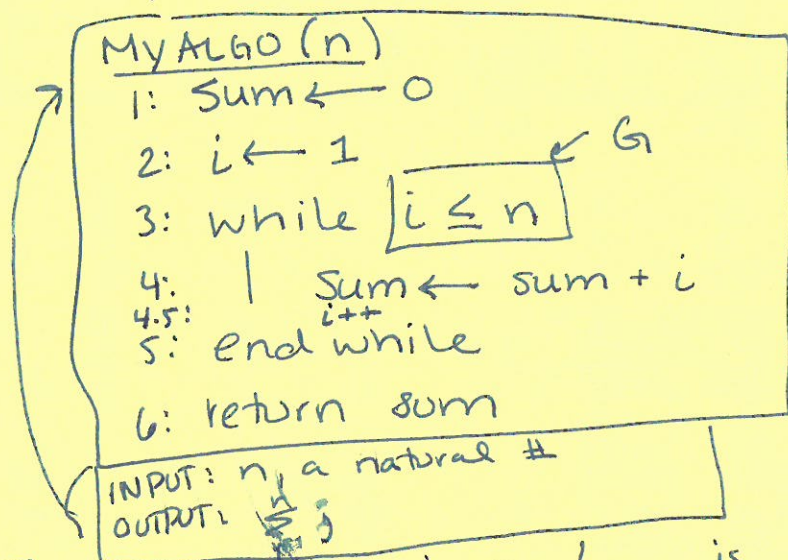


4 October 2023

LOOP INVARIANTS

The tool that we use to prove that a loop is doing what it is supposed to do.

Example: Use a loop to compute $\sum_{j=1}^n j$



↳ we can do this w/out a loop

$$\sum_{j=1}^n j = \frac{n(n+1)}{2}$$

The loop invariant L is a statement (= something that evaluate to TRUE/FALSE) that makes the following work:

① INITIALIZATION: $P \Rightarrow L$

↑
the preconditions for the loop

② MAINTENANCE

L_i = the loop inv. going into the i th iteration

G = the loop guard

$$G \wedge L_i \Rightarrow L_{i+1}$$

③ END

$$\neg G \wedge L \Rightarrow Q$$

①
↳ the post-condition of the loop / what the loop is supposed to do

For My ALGO Example:

Q = the variable sum is equal to $\sum_{i=1}^n i$.
↑ the important part: helps direct us to know where to go.

G = " $i \leq n$ " $\Rightarrow \neg G$: " $i > n$ "

↑ a low-hanging fruit b/c easy to spot

P ← the preconditions: what do we know at the time we enter line 3 (for the first time)?

P = P1: " n is a natural number"

and " $i = 1$ " P2

and " $\text{sum} = 0$ " P3

L: $\text{sum} = \sum_{j=1}^{i-1} j$, with $\sum_{\emptyset} j := 0$ } L(1)

and i is an integer } L(2)

and $i \leq n+1$ } L(3)

① INITIALIZATION

Since P is true, we know P_1 ($i=1$).

~~and~~ and P_3 ($\text{sum} = 0$).

$$\text{Since } \sum_{j=1}^{i-1} j = \sum_{j=1}^0 j = \sum_{\emptyset} j := 0,$$

$$\text{We have } \text{sum} = \sum_{j=1}^{i-1} j. \quad \square$$

Also, since $i=1$, L_2 and L_3 are satisfied too.

② END

Since $\neg G$, we have $i > n$. By L_3 , $i \leq n+1$ and $i \in \mathbb{Z}$.
Thus, $i = n+1$.

Since L is true, we know

$$\begin{aligned} \text{sum} &= \sum_{j=1}^{i-1} j \\ &= \sum_{j=1}^{(n+1)-1} j \\ &= \sum_{j=1}^n j, \end{aligned}$$

as desired. $\square$

② MAINTENANCE : $G \wedge L_i \Rightarrow L_{i+1}$

In words: If L is true and G is true (= we enter the loop), then L is still true when we exit the loop / try to re-enter the loop.

Proof of $L_{(1)}$:

Since G holds, $i \leq n$ and we enter the loop.

Since L holds, $\text{sum} = \sum_{j=1}^{i-1} j$

In Line 4, we add i to sum. So, after Line 4

$$\text{sum} = \sum_{j=1}^{i-1} j + i = \sum_{j=1}^i j.$$

In Line 4.5, i is increased by 1. ~~so~~ so, now,

$$\text{sum} = \sum_{j=1}^{\text{old-}i} j = \sum_{j=1}^{\text{new-}i-1} j = \sum_{j=1}^{i-1} j$$

Therefore, $L_{(1)}$ is now true going into the next iteration of the loop.

Exercise: Prove that $L_{(2)}$ and $L_{(3)}$ also hold after the i^{th} iteration through the loop.

In groups, write Binary search as a loop
and come up with the loop invariant.

Challenge: can you prove INIT.
MAINT.
and END?